

HAT-P-27b

R-0030

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_260425 · created 2026-07-08T17:58:42+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461155.7366 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.04
Transit depth (Rp/Rs)^2	1.92 +/- 1.11 [%]
Orbital Inclination (inc)	85.63 +/- 2.88
Airmass coefficient 1 (a1)	2.31 +/- 1.29
Airmass coefficient 2 (a2)	-0.58 +/- 0.64
Scatter in the residuals of the lightcurve fit is	52.59 %
Best Comparison Star	#3 - [214, 386]
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.071 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.04 vs 0.119 (deviation 0.5σ)
Duration fit vs published	0.071 d vs 0.07 d (deviation 0.03σ)
Mid-time O–C	-20.8 min
Depth SNR	3.5
Residual scatter	52.59 %

Light curve

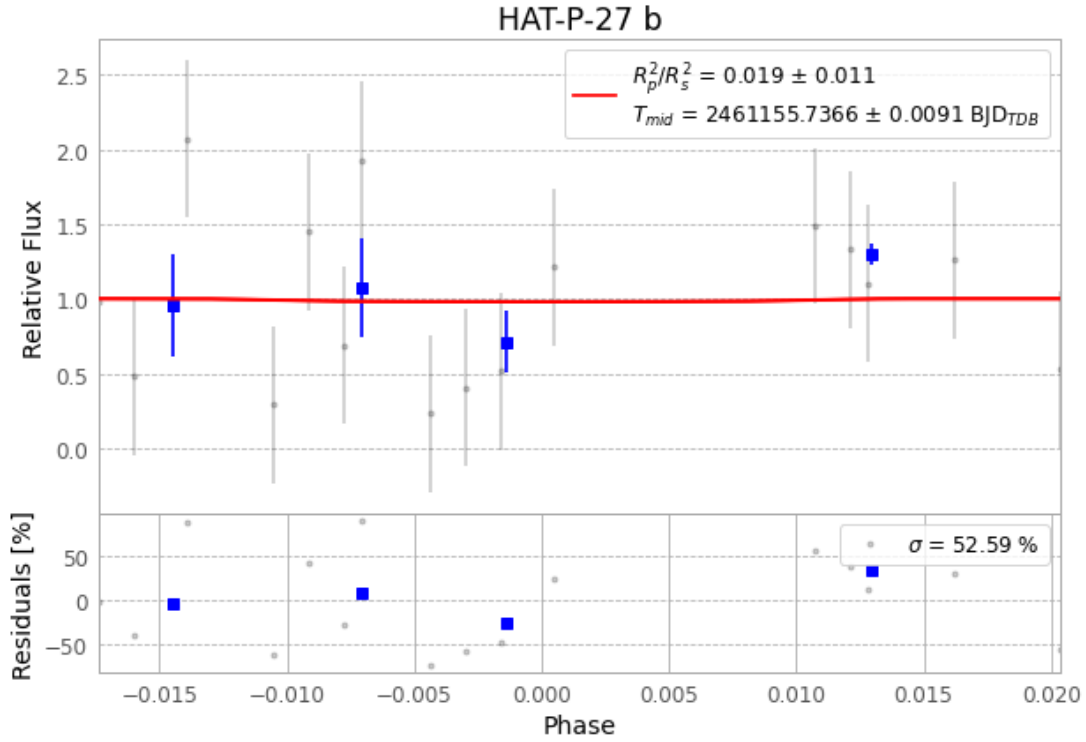


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-04-24.png (SHA-256 8110cea1b30485ee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 044938e24fa89d31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

58 FITS frames (38 MB), HATP-27260425040915.FITS → HATP-27260425070016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260425.log (d7192c05f411...), FinalLightCurve_HAT-P-27 b_2026-04-24.png (8110cea1b304...), FinalLightCurve_HAT-P-27 b_2026-04-24.csv (a46e2203a6ab...)

Compute resources

Wall time 133.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.